Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

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Today: from the population closure to an equilibrium transition

Last time: we focused on the missing population closure:

$$\text{births today} \longrightarrow \text{new households later} \longrightarrow \text{housing demand and prices.}$$

Today: I try to make that idea an equilibrium object:

- build the simplest model with a well-defined steady state;
- separate the immediate response to a fertility-preference shock from the later demographic adjustment;
- add the same population accounting to the existing quantitative model;
- ask whether the 2007–2023 economy can be interpreted as a point along such a transition.

The household problem remains the same. The update is an attempt to close the aggregate equilibrium around it.

A toy economy

To fix ideas, let's look at a very simplified economy with young and old households:

- Y_t is the number of young households at date t; O_t is the number of old households.
- A young household chooses consumption, housing services, and n_t **household children**. Next period it becomes old.
- In this unitary-household model, each household child becomes one young household in the next generation.
- Both generations purchase housing services in the same market at price p_t.

The cohort law is therefore

$$Y_{t+1} = n_t Y_t, \qquad O_{t+1} = Y_t,$$

so $n_t = 1$ is replacement.

Focus of the toy model: fertility, cohort replacement, and the housing-services market. The quantitative model adds tenure choice, mortgages, saving, and bequests.

Young households choose housing and children

A young household earns y and chooses consumption c, housing services h, and household children n:

$$u(c, h, n) = c + \alpha \log h + v \log n,$$

$$c + ph + \left[q + a(p + \omega)\right] n = y.$$

- α is the value of adult housing; v is the value of children.
- Each household child requires a units of housing and q units of other goods.
- ω is an additional cost of accessing child-suitable housing.
- Hence the resource cost of one household child is $q + a(p + \omega)$.

Old households do not choose fertility. Their reduced-form housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p},$$

where $\bar{h}_O$ measures their demand for housing services.

Housing costs jointly determine fertility and housing demand

The young household's interior choices are

$$n(p) = \frac{v}{q + a(p + \omega)}, \qquad h_Y(p) = \frac{\alpha}{p} + a n(p).$$

Thus a higher p has two effects: it reduces adult housing demand and raises the housing cost of a household child, reducing n.

Aggregate housing supply is

$$H^s(p) = \bar{H} p^{\varepsilon},$$

where $\bar{H}$ is the scale of housing supply and $\varepsilon \geq 0$ is its elasticity. At each date, p_t clears the market:

$$Y_t h_Y(p_t) + O_t h_O(p_t) = H^s(p_t).$$

This one price therefore links household fertility to aggregate cohort sizes.

A steady state

Because one household child becomes one young household,

$$Y_{t+1} = n(p_t)Y_t, \qquad O_{t+1} = Y_t.$$

A steady state is a constant triple (Y^*, O^*, p^*). It therefore requires

$$n(p^*) = 1, \qquad Y^* = O^*.$$

The fertility choice pins down the steady-state housing cost,

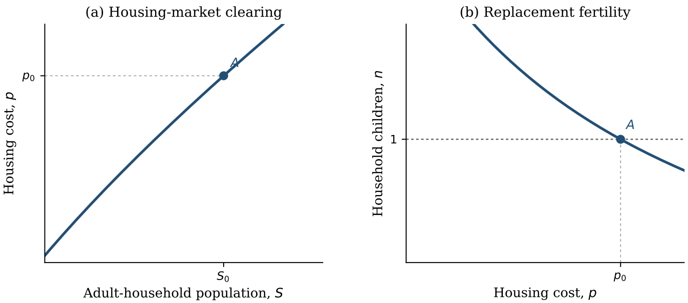
$$p^* = \frac{v - q}{a} - \omega,$$

and housing clearing pins down the population scale $S^* = Y^* + O^*$:

$$S^* = \frac{2\bar{H}(p^*)^{1+\varepsilon}}{\alpha + \bar{h}_O + ap^*}.$$

Replacement determines the price; housing clearing determines how many households live at that price.

Stage 1: the initial steady-state equilibrium



Steady-state schedule

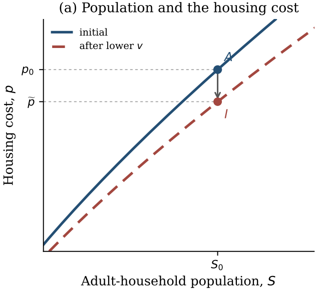
$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

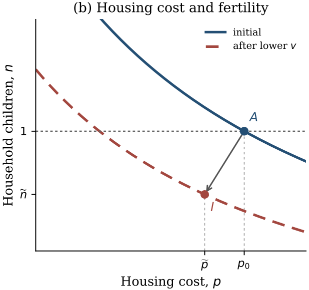
- **Replacement:** $n(p_0; v_0) = 1$ selects p_0.
- **Housing clearing:** $S_0 = S^*(p_0)$.
- **Point A:** the initial steady state.

Along the steady-state schedule, direct differentiation gives

$$\frac{dS^*}{dp} = \frac{2\bar{H}p^{\varepsilon}\left[(1+\varepsilon)(\alpha + \bar{h}_O) + \varepsilon ap\right]}{(\alpha + \bar{h}_O + ap)^2} > 0 \qquad \text{for } \varepsilon \geq 0.$$

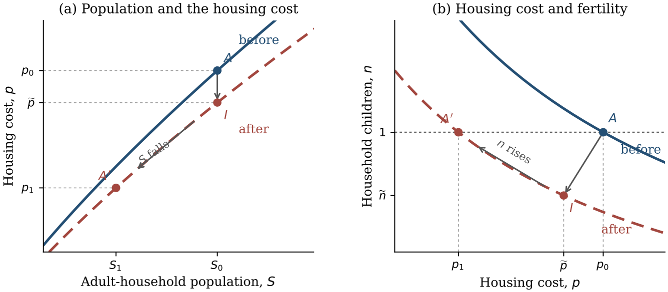
Stage 2: the impact response





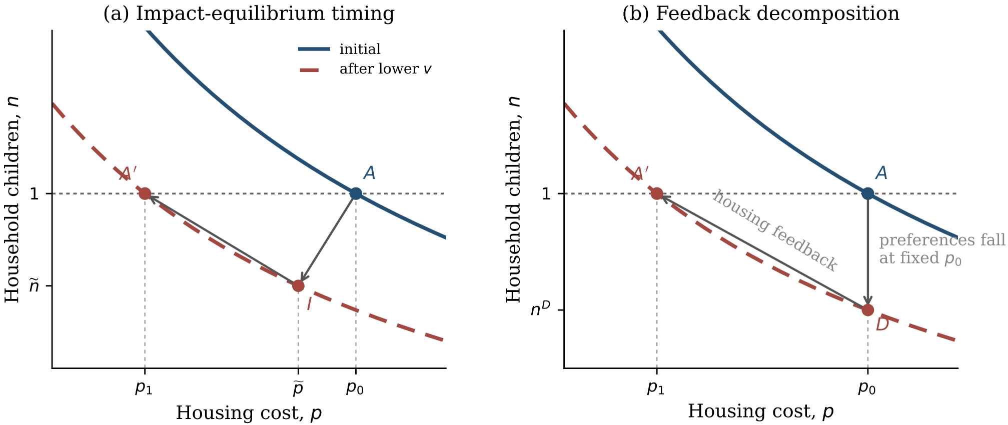
- **Shock:** $v_0 \to v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$, $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives $p_0 \to \widetilde{p}$.
- **Fertility:** $\widetilde{n} = n(\widetilde{p}; v_1) < 1$.
- **Point I:** the impact equilibrium with inherited cohort sizes.

Stage 3: demographic adjustment



- **Below replacement at I:**
$$Y_{t+1} = \tilde{n} Y_t < Y_t.$$
- **Cohorts:** smaller young cohorts eventually reduce S.
- **Housing cost:** lower demand moves $\tilde{p} \to p_1$.
- **Fertility:** lower p raises n along the red curve.
- **Point A':** $n = 1$, $p_1 < p_0$, and $S_1 < S_0$.

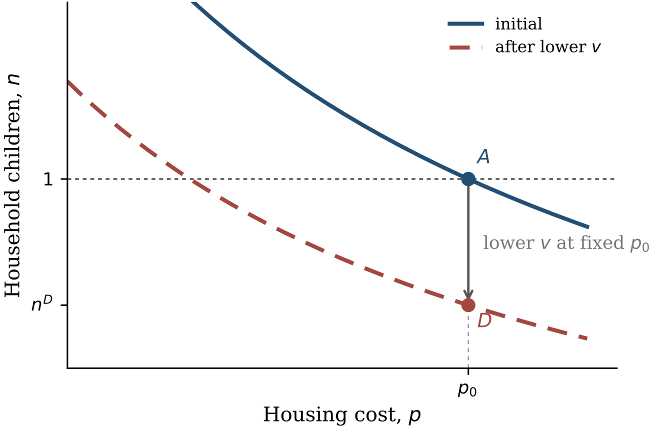
Two ways to organize the same feedback



Impact timing: I is the market-clearing impact point with inherited cohorts.

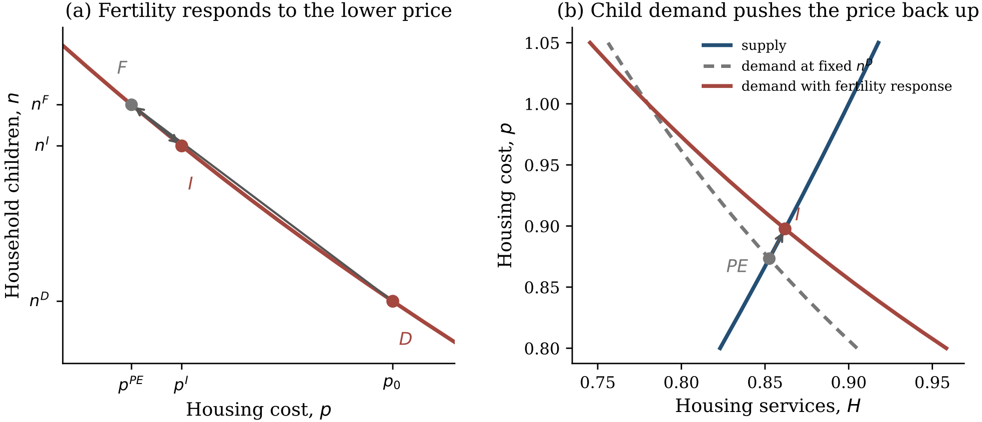
Feedback decomposition: D holds $p = p_0$; the move to A' shows how falling housing costs prevent the no-feedback demographic contraction from continuing.

Alternative 1: the primitive fertility shock



- **Primitive:** $v_0 \to v_1 < v_0$.
- **Hold** $p = p_0$**:** fertility falls from 1 to n^D.
- **Point** D**:** prices and cohort sizes have not yet adjusted.

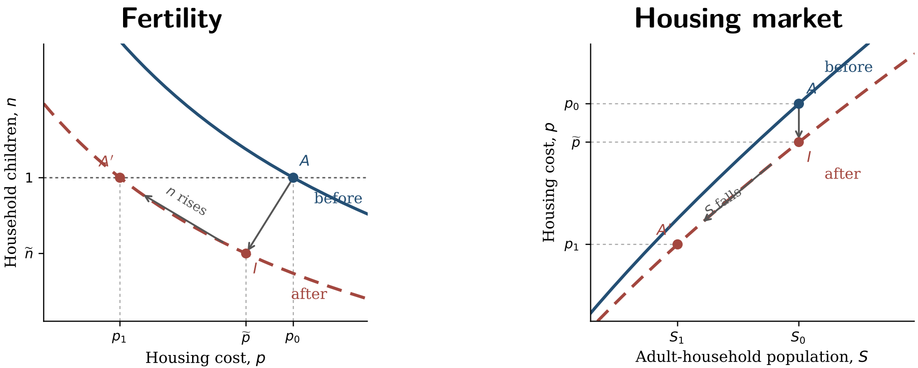
Alternative 2: fast equilibrium feedback



Holding $n = n^D$ gives the mechanical price p^{PE}. At that price households want $n^F > n^D$, so child-related demand pushes the price back to p^I and fertility down to n^I:

$$p^{PE} < p^I < p_0, \qquad n^D < n^I < n^F < 1.$$

Alternative 3: slow cohort adjustment



- At I, $n^I < 1$: smaller cohorts gradually reduce housing demand and p.
- Lower p raises fertility along the new curve until the economy reaches A' with $n = 1$ and $S_1 < S_0$.

Below-replacement fertility need not lower population immediately

Let $n_t = n(p_t)$ be household children per young household and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $n_t < 1$ whenever

$$\mu_t < n_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \quad \longrightarrow \quad 80 \text{ young} + 100 \text{ old}$$

$n_t = 0.8 < 1$, but total households rise from 170 to 180.

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1-\mu_t)h_O(p_t) > (1-n_t)h_Y(p_t).$$

U.S. interpretation: inherited cohorts can keep population and housing costs rising even after fertility falls below replacement.

Bringing the closure into the quantitative model

Now return to the quantitative lifecycle model. At each date, households differ in age, wealth, income, location, tenure, house size, fertility history, and children at home.

Given these characteristics, households choose fertility, tenure, housing, saving, and moving. These choices determine both their own next-period circumstances and the number of future entrant households.

To study the aggregate transition, we carry the entire household distribution forward in calendar time:

$$\text{households at date } t \longrightarrow \text{choices, survival, and births} \longrightarrow \text{households at date } t+1.$$

At every date, this distribution determines births, housing demand, and the market-clearing housing cost. Births then feed back into the distribution when children form their own households.

The quantitative transition embeds the existing lifecycle decisions inside the population–housing feedback of the toy model.

Three equations close the quantitative transition

1. **Births generate future households.** Adjusted births b_t at date t become entrant households twenty years later:
$$E_{t+5} = \frac{b_t}{2.1}.$$
One model period is four years; the factor 2.1 maps children into future household units.

2. **The household distribution evolves in levels.** Let G_t denote the full distribution across age and household characteristics. Then
$$G_{t+1} = \text{surviving households after their choices} + \text{new entrants}.$$

3. **The housing market clears at every date.**
$$H^s(p_t) = D(G_t, p_t),$$
where $D(G_t, p_t)$ is total housing demand generated by the households in G_t.

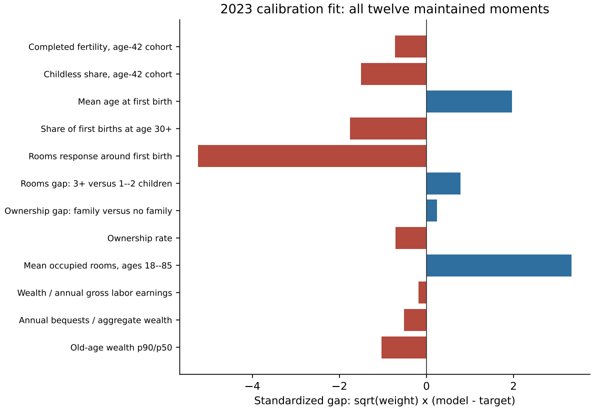
Housing costs shape fertility and housing choices; those choices change future cohort masses; future cohorts feed back into housing costs.

We calibrate an economy moving from 2007 to 2023

1. **Construct an old reference economy.** Choose the initial fertility preference so completed fertility is 2.1. This is a replacement normalization, not a measured 2007 target.
2. **Construct the 2007 state.** Reweight the old distribution only by observed 2007 age shares, preserving wealth, tenure, and fertility states within each age.
3. **Run the economy to 2023.** Feed in Census household totals and ACS age shares. Let the fertility preference decline linearly, and estimate the household parameters and total decline using twelve maintained moments, evaluated mainly in 2023.
4. **Check the path.** Compare untargeted model movements with period fertility, birth timing, ownership, rooms, house prices, and rents.
5. **Continue after 2023.** Hold preferences fixed and let the inherited household distribution and birth pipeline evolve.

The population and age bridge are inputs, not successes of the model. The 2007 economy is a dated initial condition, not literally the old steady state.

The 2023 fit has two important housing-size misses



Source: final dated-transition calibration; selected candidate 7.

Calibration facts

Loss: 50.742**.** Ten parameters and twelve moments.

Two exact independent repeats.

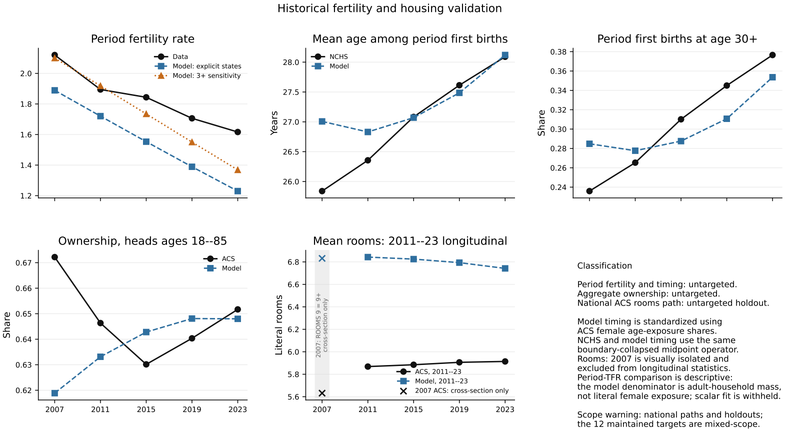
Main misses

- First-birth rooms: 0.273 versus 0.720
- Mean rooms: 6.742 versus 5.780

They contribute 38.57 of the total loss.

The PSID rooms target is provisional because its prepath is not flat.

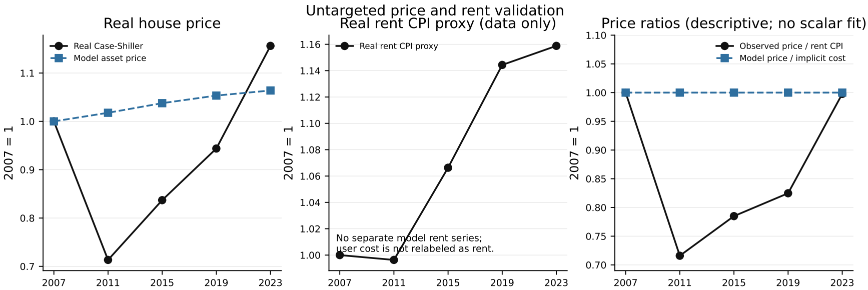
Historical validation is mixed rather than hidden



Reading the figure: timing is close by 2023; the model misses the ownership cycle and rooms trend, and its fertility decline is too steep.

National series are holdouts; maintained targets use mixed samples and vintages.

The model does not reproduce the house-price cycle



- The model cannot reproduce the 2007–2011 house-price collapse and later rebound.
- With temporary-equilibrium expectations and a fixed user-cost relation, it has no independent price-to-rent dynamics.

The exercise should therefore be read as a demographic-housing transition, not as a complete account of the post-2007 housing cycle.

Observed house prices: real Case-Shiller. The rent CPI is shown only as a descriptive proxy, not as the model's user cost.

What exactly is the post-2023 continuation?

Both exercises begin from the **same calibrated 2023 distribution**: the same ages, wealth, tenure, fertility histories, children at home, and birth pipeline. We then hold the 2023 fertility preference fixed and introduce no policy.

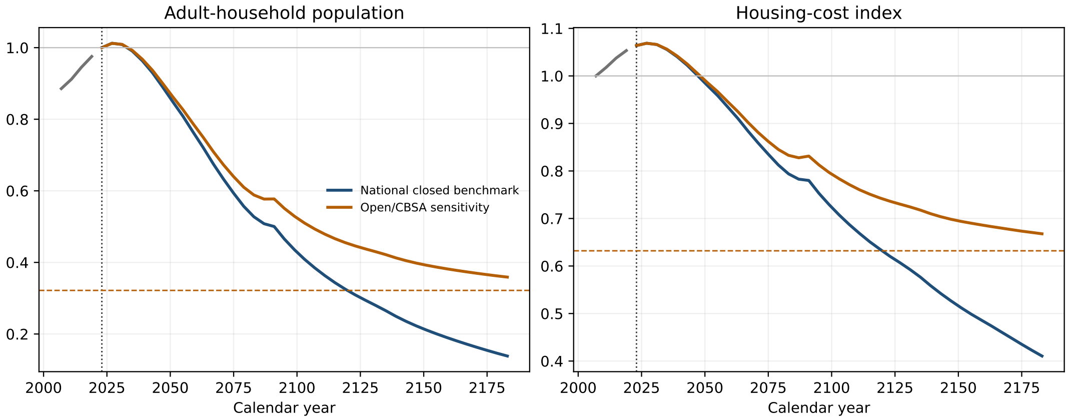
The two paths differ only in how new households enter after 2023:

- **Closed benchmark.** There is no outside entry. Future entrants come only from earlier birth cohorts.
- **Open sensitivity.** A fixed outside flow M and a fixed local-retention rate ρ continue from the old reference economy.

Because births take twenty years to become entrant households, the first post-2023 years are largely predetermined. Inherited cohorts can make population and housing costs rise briefly even though reproduction is below replacement.

These are controlled equilibrium experiments, not forecasts. The open case does not hold the outside-origin entrant *share* fixed: 0.169 only normalizes the old reference state.

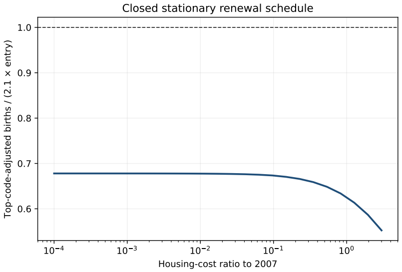
Demographic momentum delays, but does not prevent, decline



Both paths peak slightly in 2027 because the initial age structure and birth pipeline are inherited. Afterward, the closed economy contracts much faster. By 2183, population is 0.138 of its 2023 level in the closed benchmark and 0.359 in the open sensitivity; asset prices are 0.386 and 0.628.

In the open stationary endpoint, the realized outside-origin entrant share is 0.465 because the fixed flow M becomes large relative to a shrinking population.

The fitted closed economy has no positive terminal steady state



25-point grid, $P/P_0 \in [10^{-4}, 3]$; not a global theorem.

At each stationary price P, let $E(P)$ be entrants and $B(P)$ the future households generated by births. A positive closed steady state requires $B(P)/E(P) = 1$.

Instead, over the audited grid,

$$0.552 \leq B(P)/E(P) \leq 0.678 < 1.$$

Even almost free housing does not reproduce the entrant cohort.

Therefore: the closed continuation is a finite-horizon experiment, not a transition between two positive steady states.

Where this leaves the paper

- **The theoretical contribution is compact.** The existing housing–fertility mechanism is embedded in a population closure. Lower fertility eventually reduces population and housing costs, while inherited cohorts can make both rise initially.
- **The quantitative exercise is a dated transition.** We initialize in 2007, fit the maintained 2023 moments, and keep imposed demographic inputs separate from historical holdouts.
- **The present fit is informative but incomplete.** Several 2023 moments fit well, but housing quantities, the ownership cycle, and the house-price path do not.
- **The equilibrium problem is substantive.** The current fertility response is too weak to restore replacement at any audited positive price, so the closed continuation has no positive terminal steady state.

The immediate decision is narrow: either present a finite closed benchmark with an explicit open sensitivity, or re-estimate the housing-cost-to-fertility slope under a closed-root requirement before claiming a transition between steady states.

Policy analysis comes only after this equilibrium choice is settled.